A Finite Law-Control Obstruction for the $c = 1$ Prime-Shell Band

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Abstract

TPC-378 transferred a finite $c = 1$ band-support profile to three fresh coordinate-disjoint origins. This paper tests whether the profile is a property of the common mask and square-energy geometry, or of the all-plus prime-shell law. We compare four predeclared laws on a new affine-grid panel. The complete panel has 36 rows with $N = 1024$ and $Q = 512, 2048, 8192$. The all-plus law has profile $(0, 3, 3)$, while every signed control has profile $(0, 0, 0)$; there are 6 spectral-cap failures and no Schur-cap failures. This is a finite sign-law-dependence obstruction, not an arithmetic or twin-prime theorem.

1 Question and claim boundary

The preceding finite experiments used the all-plus shell sum

$$A(u,t) = \sum_{Q<p\le 2Q} K_p(u,t).$$

Its normalized near-block band exceeded the working spectral cap at the two larger Q anchors. The present experiment holds the geometry, band, origins, count, and Q ladder fixed while changing only a predeclared sign law.

All conclusions concern one explicit finite panel. A signed control is a diagnostic sign sequence, not a character sum known to represent the twin-prime source. The terms `OPEN`, `MODELING_CHOICE_OPEN`, and `NO` in the project firewall are part of the result.

2 Finite operator and four laws

For $I = [a, a+N-1] \cap \mathbb{Z}$, $p \in (Q, 2Q]$, and $u, t \in I$, put

$$K_p(u,t) = p\left(\frac{p}{Q}\right)^2 \frac{66^2}{66^2 + (u-t)^2}\left(\mathbf{1}_{p\mid(u-t)} - \frac{1}{p-1}\right)\mathbf{1}_{u\ne t}\mathbf{1}_{p\nmid u}\mathbf{1}_{p\nmid t}.$$

For a law ℓ with signs $s_p(\ell) \in \{-1, 1\}$, define

$$A_\ell(u,t) = \sum_{Q<p\le 2Q} s_p(\ell) K_p(u,t), \qquad G(u) = \sum_{t\in I}\sum_{Q<p\le 2Q} K_p(u,t)^2,$$

$$T_\ell(u,t) = \frac{A_\ell(u,t)}{\sqrt{G(u)G(t)}}.$$

The geometry G is common to all four laws. The signs are fixed in the ordered prime shell as follows:

ℓ	$s_p(\ell)$
all_plus	1
alternating_index	$(-1)^{\mathrm{index}(p)}$
mod4_character	1 if $p \equiv 1 \pmod 4$, -1 otherwise
half_split	1 on the first shell half, -1 on the second

For $b(u) = \lfloor (u-a)/256 \rfloor$, use the common band and tail

$$B_\ell(u,t) = T_\ell(u,t)\mathbf{1}_{|b(u)-b(t)|\le 1}, \qquad R_\ell = T_\ell - B_\ell.$$

For any unit eigenvector v of T_ℓ with eigenvalue λ,

$$v^{\mathsf T} B_\ell v + v^{\mathsf T} R_\ell v = \lambda$$

is an exact finite identity.

3 Predeclared protocol

Before any metric is read, fix

$$a_j = 1200001 + 401j, \quad 0 \le j < 41,$$

and select indices $(0, 20, 40)$, giving origins $(1200001, 1208021, 1216041)$. Use $N = 1024$, four contiguous blocks of length 256, exponent one, beta 2, height 66, and band cutoff one. The complete $3 \times 3 \times 4 = 36$ Cartesian panel uses $Q = 512, 2048, 8192$, with spectral and Schur caps 0.64 and 0.83. No origin, law, row, or failure census is selected from a response.

The current intervals are disjoint from the largest declared intervals in TPC-376–378 by exact endpoint inequalities. The exact anchor $[1200001, 1200014)$ at $Q = 8$ has shell $\{11, 13\}$; rational arithmetic checks positive common geometry and symmetry for every law. Its geometry digest is `4436feb8f2abf5450599e5d9185c28e245f07c9829000812e8ab0eb18726eb86`.

4 Results

Table 1 reports band spectral-cap failures among the three origins at each Q.

Table 1: Finite law-control panel.

law	$Q = 512$	$Q = 2048$	$Q = 8192$	max spectral	max Schur
all_plus	0/3	3/3	3/3	0.653347588	0.679884719
alternating_index	0/3	0/3	0/3	0.009408454	0.027602425
mod4_character	0/3	0/3	0/3	0.011835976	0.026614970
half_split	0/3	0/3	0/3	0.211734949	0.222094649

Thus the all-plus profile is $(0, 3, 3)$ and each signed control has profile $(0, 0, 0)$. The total spectral census is 6/36 and the Schur census is 0/36. The largest band spectral values in the displayed law order are 0.65334758792533143, 0.0094084540584888146, 0.011835976723613296, and 0.2117349490215118. The selected full-mode band-Rayleigh absolute retention over all rows ranges from 0.0045649378279021191 to 0.98047576299453554; the maximum absolute tail fraction is 0.99543506217209821.

5 Audits and interpretation

The producer writes a canonical JSON certificate and locks the TPC-378 source and certificate. An independent checker uses a direct sieve through 20000, reverse-shell accumulation, independent sign patterns, common geometry, and independent full and band eigensystems without importing the TPC-379 producer. Normal and optimized runs are byte-identical. A 25-mutation stress suite changes protocol, law, row, census, and firewall fields; every mutation is rejected. Local Bridge-B repeats these checks and locks the stable project artifacts and diagnostic-free compile log.

The strongest positive result is a complete finite, response-blind, coordinate-disjoint comparison on one common geometry. The strongest obstruction is that the all-plus high-Q signature disappears under all three signed controls. It therefore cannot be promoted as a law-invariant property of the finite mask; this is not a theorem choosing a law.

6 Route status and conclusion

The official Session-named Route-A and Route-B evaluator files are absent from this checkout. The project route note and local Bridge-B are fail-closed repository evidence rather than an official evaluator verdict. The exact finite protocol, coordinate separation, common geometry, sign definitions, and band/tail identity are proved finite statements. The replay and census are numerically certified finite statements.

Source-validity of the normalization, law/origin/scale uniformity, cross-block causality, a growing operator bound, source-uniform L^2, and signed prime-shell reassembly remain open. The certificate records

`ARITHMETIC_ADVANCE=NO`, `FIXED_POWER_CREDIT=0`, `FULL_GATE_B=OPEN`,

with no twin-prime conclusion. The next minimal finite test is `TEST_C1_LAW_CONTROL_COUNT_REPLAY`: repeat the frozen law family at count 2048 on a new response-blind origin panel.